UDIT JAIN

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Agentic AI Engineer | LLM Post-Training & RL | Samsung Research AI Core Team

PROFESSIONAL EXPERIENCE

Samsung Research — AI Core Team Seoul, South Korea
Agentic AI Engineer *January 2026 – Present*

- End-to-end LLM post-training (SFT, RLHF, DPO) and RL (PPO, reward modeling, process/outcome reward signals) for instruction following, safety alignment, multi-step reasoning, and autonomous agentic task completion.
- Knowledge distillation from frontier models to deployable students maintaining benchmark parity at reduced compute and latency. Built multi-turn agentic evaluation frameworks for tool use, planning, and long-context reasoning.
- Distributed training infrastructure with NeMo on multi-node GPU clusters; large-scale inference and serving with vLLM.

Samsung Research — Next Gen. Digital Appliances R&D Seoul, South Korea
Computer Vision Engineer *January 2023 – December 2025*

- Developed and deployed vision-based AI for millions of Samsung smart refrigerators (TensorFlow, PyTorch, OpenCV) — real-time food recognition, inventory management, and personalized dietary recommendations in production.
- Core CV: object detection, instance segmentation, feature matching, trajectory analysis, gaze prediction, and HAR.
- +4% accuracy via OOD detection (openset loss, contrastive learning, ensembles). NPU: <40MB, <70ms, >90%. CES 2024.

Samsung Research — Data Service Lab Seoul, South Korea
Software Engineer *January 2022 – December 2022*

- Samsung Health Research Application — connecting Samsung devices with clinical research at Samsung Medical Center. Full-stack: Kotlin/Android, Node.js, Apache Superset. Won UX Design Award 2023.
- Managed IRB-compliant data collection connecting Samsung wearables to clinical research protocols for cardiac monitoring, diabetes management, and sleep disorder studies at Samsung Medical Center and partner hospitals.

Samsung Research — Data Intelligence Lab Seoul, South Korea
Data Science Engineer *October 2020 – December 2021*

- Large-scale analysis on Samsung Health's 200M+ user database (weight, BMI, sleep, exercise) via AWS EMR, Athena, Spark, BigQuery. Secure data pipelines for 300TB health data on AWS S3. Sleep study featured on Korean national TV.
- Co-authored pandemic-era global sleep study published on Samsung Newsroom, analyzing behavioral and circadian rhythm shifts across millions of Samsung Health users in 20+ countries.

EDUCATION

Indian Institute of Technology Delhi (IIT Delhi) New Delhi, India
B.Tech in Computer Science and Engineering — CGPA: 9.10/10.0 *2016 – 2020*

Coursework: Computer Vision, Machine Learning, AI, Deep Learning, Computer Graphics, Operating Systems, Algorithms

RESEARCH & INTERNSHIPS

Harvard University — Visual Computing Group Massachusetts, USA (Remote)
Research Intern *March 2020 – August 2020*

- Optimized PyRedner differentiable rendering library for inverse rendering; GPU-accelerated Monte Carlo simulation.

Samsung Research — Data Analytics Lab Seoul, South Korea
Research Intern *May 2019 – July 2019*

- Multilingual NLP system for semantic extraction from B2C communications (SpaCy, NLTK, OpenIE, StanfordIE).

Kyutech Institute of Technology — Shibata Labs Fukuoka, Japan
Research Intern *May 2018 – July 2018*

- Elderly care robotics — Mask R-CNN cloth segmentation + Baxter robot control via Kinect. RoboMech 2018, iREX 2019.

PUBLICATIONS

APEX: Adaptive Ext4 File System for Enhanced Data Recoverability in Edge Devices — IEEE Cloud Computing, Sydney 2019. arXiv:1910.01642
Transformative Effects of IoT, Blockchain and AI on Cloud Computing — cs.DC, 2019. arXiv:1911.01941

TECHNICAL SKILLS

LLM & Post-Training: RLHF, DPO, PPO, SFT, reward modeling, process reward models, knowledge distillation, PEFT/LoRA
Frameworks: NeMo, vLLM, PyTorch, TensorFlow, HF Transformers, OpenCV, TensorFlow Lite
Languages & Cloud: Python, C/C++, Kotlin, Java, SQL, Node.js | AWS (S3, EMR, Glue, Athena), BigQuery, Spark
Infrastructure: Distributed GPU training, Docker, Linux, NPU optimization, FPGA | CV, NLP, robotics, edge AI

SELECTED ACHIEVEMENTS

- KVPY Fellowship — All India Rank 3 / 500,000 students (IISc Bangalore, 2015)
- JEE Advanced — All India Rank 81 / 200,000 qualified candidates (2016)
- IIT Delhi Institute Merit Award — top 7% of batch (2017-18)
- Best Paper Award — Samsung Electronics Research Division (2022)
- Samsung Certifications — AI Development (2021), Cloud Architecture (2021), Data Science L2 (2023)
- Korean Language — SNU Level 3-4 Certificate of Excellence (2023, 2024)
- National Science Olympiads — Physics Gold Medal (HBCSE); qualified Physics, Chemistry, Astronomy